

		unit of time	s
3	B	Volume of optical fibre = cross-section area $\times$ length $= \frac{\pi}{4} d^2 \times L$ $= \frac{\pi}{4} (250 \times 10^{-6})^2 (50 \times 10^3)$ $= 2.45 \times 10^{-3}$ $\approx 2.5 \times 10^{-3} \text{ m}^3$	
4	C	1	